

JACOB DORRILL

Machine Learning Engineer

083 376 4132 | jacobdorrill@gmail.com

jacobdrizzle.dev | [GitHub](#)

PROFESSIONAL PROFILE

ML Engineer currently building predictive maintenance models, OCR pipelines, and RAG systems at CompuCal. Day-to-day work involves training and benchmarking gradient boosting and deep learning models on calibration data, embedding documents into pgvector for retrieval, and wiring AI agents to production databases via MCP. Python-first, comfortable with AWS, and experienced across the pipeline from raw data to deployed API.

TECHNICAL SKILLS

ML & AI	PyTorch, TensorFlow, Scikit-learn, CatBoost, XGBoost, LightGBM, CrewAI, LangChain, Pandas, NumPy
GenAI & NLP	RAG Pipelines, Prompt Engineering, pgvector, Embeddings, Azure Document Intelligence, GLM-OCR, MCP
Languages	Python (primary), JavaScript, TypeScript, Java, C++, Rust, Kotlin
Cloud & DevOps	AWS (S3, EC2, Lambda), GCP, Docker, Kubernetes, CloudFlare, Vercel
Databases	PostgreSQL, pgvector, Snowflake (familiar), MySQL, MongoDB, Neo4j, Firebase, GraphQL
Backend	FastAPI, Flask, Django, Node.js, Express.js, REST API Design
Frontend	React, Next.js, Tailwind CSS
Tools	Git, GitHub, Linux (primary OS), VS Code, JetBrains IDEs

PROFESSIONAL EXPERIENCE

CompuCal | Cork, Ireland Current
Machine Learning & FullStack Engineer (Intern)

Taking a legacy calibration management platform and adding ML-driven features: predictive models, OCR extraction, RAG chatbot, and agent-based automation.

- Predictive Maintenance ML:** Built and benchmarked classification, regression, and survival models (XGBoost, LightGBM, CatBoost, Random Forest, Cox PH, LSTM, TabNet) on a 557K-row multi-customer calibration dataset to forecast instrument failures and recommend calibration intervals; top model reached ROC AUC 0.94 / F1 0.74.
- OCR Pipeline:** Built a multi-vendor OCR extraction service for calibration certificates using Azure Document Intelligence and a self-hosted GLM-4V vision model (vLLM on Azure); deployed via FastAPI + Docker, covering 24 vendors across four languages and writing structured output directly into the calibration database.
- RAG Support Chatbot:** Shipped a dual-mode chatbot (embedded widget + full chat UI) over the product user manual using pgvector, OpenAI embeddings with an Ollama fallback, and a Next.js + FastAPI stack with streaming responses; deployed on Railway.
- MCP Server & Deep Agent Platform:** Built an MCP server exposing ML tools to Claude Code and a multi-provider orchestration layer (Claude Code CLI, Codex CLI, OpenAI API) that drives an automated Jira → code → review → PR pipeline, with a Next.js dashboard for live run monitoring.
- Document-Drift Automation:** FastAPI agent that reviews pull-request diffs against functional specs and auto-commits documentation updates — critical in a regulated calibration environment (ANSI/NCSL Z540.3, ISO 17025).
- Developer Tooling:** Rolled out Claude Code across the dev team with custom skills, agents, and hooks tuned to the ML pipeline, contributing to ~30% throughput gain.
- Whitepaper & Executive Strategy:** Authored the CompuCal ML Whitepaper and an executive pitch deck outlining an AI-Native Calibration Enterprise strategy.

TECHNICAL PROJECTS

ReviveMap | Google Hackathon — 3rd Place

Python, Gemini Flash, Next.js

AI-powered platform for identifying and analysing derelict buildings in Ireland to compute investment viability.

- **Rule Engine:** Python orchestration layer that pulls property records from Firestore, fans out parallel enrichment tasks (geocoding, image analysis, grant lookup), and writes scored results back.
- **Vision Extraction:** Fed property images to Gemini 2.5 Flash with structured output schemas; extracted renovation cost breakdowns, material types, and condition ratings as typed JSON.
- **Scoring:** Weighted composite scores (Viability, Community Value, Sustainability) from air quality indices, amenity proximity, BER ratings, and SEAI/Vacant Property Grant eligibility data.
- **Data Flow:** Scraper → Firestore → Python Rule Engine → Next.js frontend. Images stored in Cloudflare R2 with signed URLs.

AI-Powered Stock Market Prediction | Agile Module Project

Python, Pandas, FinBERT (Hugging Face), XGBoost, Scikit-learn

- **Sentiment Features:** Ran Federal Reserve transcripts through FinBERT at sentence level to extract sentiment scores from nuanced financial language, then used these as input features for downstream models.
- **Prediction Models:** Trained XGBoost classifiers to forecast both direction (Up/Down) and magnitude of S&P 500 moves in the 24 hours following Fed announcements.
- **Backtesting:** Event-driven backtest on a hold-out dataset simulating a trading strategy with strict temporal splits to prevent lookahead bias. Evaluated both model accuracy and financial return.
- **Process:** Agile delivery with sprint planning, daily stand-ups, and retrospectives across the project lifecycle.

NeuroMind System (NMS)

Python, CrewAI, React, REST API

Multi-agent healthcare platform for real-time patient risk assessment and cognitive monitoring.

- **Agent Backend:** Python backend using CrewAI where separate agents handle patient data ingestion, risk score calculation, and medical news summarisation. Agents run autonomously and write results to a shared state store.
- **API Layer:** REST API serving as the sync point between Kotlin Android clients, React web dashboard, and the inference engine. Handles auth, role-based access, and real-time data propagation.
- **Clinician Dashboard:** React portal with role-based views, cohort tracking tables, and time-series visualisations of cognitive decline metrics per patient.
- **Mobile Data Collection:** Kotlin modules capturing speech recordings and cognitive test inputs on-device, batching and uploading to the Python backend for model inference.

EDUCATION

BSc in Software Development

2024 – 2026

MTU Cork | 3rd Year | 78.55% GPA

Student Inc. Business Incubator

2025 (May – Aug)

Rubicon Centre, Cork | Grade: 84% | Business strategy, marketing, finance, IP

QQI Level 6 in Software Development

2023 – 2024

Morrisons Island Campus | Distinction | OOP (C++), Software Architecture, Web Development

Earlier:

New Frontiers Entrepreneur Programme Phase 1 | 2023

QQI Level 5: Kick Start Your Own Business | 2023

QQI Level 5: IT Specialist JavaScript | 2022

QQI Level 5: Introduction to programming HTML + CSS | 2021

QQI Level 5: Organic Horticulture and Bee Keeping (2018 - 2019)